

Why music psychology needs larger and more diverse datasets and how citizen science can help

Courtney B. Hilton¹

¹School of Psychology, University of Auckland, Auckland, 1010, New Zealand.

*Email: courtney.hilton@auckland.ac.nz

Abstract: To produce more nuanced, generalisable, and culturally representative understanding, music psychology needs to engage with more diverse forms of music and study the musical experiences, perceptions, and behaviour of more diverse people. But the logistical challenges of how to feasibly collect larger and more diverse datasets can be a significant barrier to achieving this goal. While there are also several other challenges inherent to broadening the empirical scope of music psychology in this way, here, I focus on the reasons why more generalisable empirical inferences require larger and more diverse datasets. Then to meet the logistical challenge of collecting such expanded datasets, I suggest that a citizen science approach to research participation may be particularly promising, and review recent studies taking this approach. Citizen science projects can function at many different scales, but at its extremes, it can unlock a shift from studying tens of students at a local campus to studying millions of people around the world. Despite its own limitations and challenges, citizen science has the potential to play a unique role in expanding and diversifying the empirical scope of music psychology.

Introduction

Empirical studies in music psychology are primarily based on observations of how particular people respond to particular musical stimuli in particular contexts. Such observations can inform our understanding of musical behaviour and its psychological underpinnings, such as how people respond emotionally to different harmonies, how musical performers move their bodies while performing, or which brain areas are involved in processing particular musical structures. Yet in the majority of cases in the current literature, these observations are drawn from small and biased samples of musical examples, music makers, and music listeners, representing only a fraction of the musical and psychological diversity that exists around the world. While such narrow datasets may be still useful for answering questions about specific styles of music, or about specific listeners or makers of music, they are insufficient for making the far broader kinds of claims that they are often, explicitly or implicitly, put to work for. Biased and unreliable understandings are therefore likely to prevail and this has both theoretical and practical implications.

To support broader and more reliable findings, music psychology needs to study music more diversely and at scale. But increasing diversity and scale is logistically challenging. It is easier to study a few people who live nearby, share the same language and culture, and have enough interest and free time to participate in research studies. Increasing diversity requires specialised expertise, additional resources, and engagement with new ethical considerations (Urassa et al., 2021; Broesch et al., 2020; Jacoby, Margulis, et al., 2020; Woods, 2023). Scaling up these efforts to attain both large and diverse samples further compounds these logistical difficulties. Thus while researchers may wish to broaden the diversity and scale of their empirical investigations, doing so may not be practically feasible.

These logistical challenges are vital to address because what is practical is ultimately what gets done. Further, if only the most well-resourced music psychology labs are able to obtain larger and more diverse datasets, and thus to contribute to important progress, then this would introduce further inequalities into an already unequal field. So for these reasons, if indeed we should collect larger and more diverse datasets, then we should also invest serious efforts to make collecting such datasets easier for everybody. To begin, I will now review the reasoning for the first part of this argument about why it is we need larger and more diverse datasets in the first place.

Why do we need larger and more diverse datasets?

In the introduction to the first edition of *Empirical Musicology*, Clarke and Cook remarked that musicologists are “prone to build interpretations on very small data sets or even on single instances” (p. 4, [Clarke & Cook, 2004](#)). In part, they argued, this is a result of some areas of musicology being focused on topics for which data are hard to come by. David Huron calls these ‘data poor’ fields ([Huron, 1999](#)). For example, studying everyday musical experiences in 16th century Europe is hard to do empirically because there simply is not much surviving data that directly speaks to this question. Musical behaviour, for the most part, is ephemeral, leaving behind few enduring traces of its existence besides artefacts like manuscripts and instruments. While the latter do present important empirical targets for historical research, they place severe constraints on what can realistically be studied. So in distinguishing empirical musicology from other parts of musicology, Clarke and Cook emphasised engagement in topics for which abundant data are at least in principle accessible, and the use of appropriate methods for collecting and analysing such data.

The psychological sciences tend to be less historically focused, at least empirically (although there are renewed efforts to change this: [Atari & Henrich, 2023](#)). This is also true of music psychology. As a result, it is a relatively ‘data rich’ field of study, meaning that for the kinds of research questions that it poses there is an abundance of potentially relevant empirical observations that can feasibly be made. Indeed we are at almost all times surrounded by human beings who have psychologies and engage in musical behaviour of one kind or another. And if not other people, our own subjective experience furnishes us with uniquely direct observations on what it is like to perceive, experience, and think about music, at least from a biased individual perspective. However this gross abundance of empirical riches may have also bred complacency. There are approximately eight billion people around the world who could in principle participate in music psychology studies. There are also more than 100 million music recordings accessible on popular streaming platforms, that in principle could be used as stimuli in experiments. Yet the vast majority of studies collect data from mere tens of research participants, typically students from the researcher’s home campus, and use an exceedingly narrow selection of musical examples as stimuli. For all the data riches that music psychology has available, it tends to sample from only a fraction.

Is this empirical smallness a problem? It depends. At the level of an individual research study, there’s nothing inherently wrong with studying even a single musical stimulus or only a single research participant. It depends on how the results are framed and interpreted. For example, Roger Shepard’s initial paper on what has since become known as the ‘Shepard Tone Illusion’ describes a novel approach to synthesising a tone’s timbre that has striking effects on pitch perception ([Shepard, 1964](#)). This paper is none the poorer for having only compared such novel ‘Shepard Tones’ to the standard synthetic ‘puretone’ timbre. It would have been superfluous to also include a variety of other synthesised and natural timbres as comparisons. Indeed even the inclusion of the ‘puretone’ pitch condition was arguably not necessary to demonstrate the core point that ‘Shepard Tones’ violate the commonsense expectation that the greater the pitch interval between two tones, the clearer it is which tone is higher in pitch and which is lower (under more normal conditions, this assumption holds robustly for people all around the world: [Hilton et al., 2021](#)). Empirical smallness in this case is perfectly adequate to support the correspondingly specific interpretation that Shepard makes: that there are some situations in which human perception of pitch is not well explained as being linear (although c.f., [Burns, 1981](#)).

So the first point to be made here is not that every empirical research study in music psychology needs thousands of diverse participants and a dazzling array of varied stimuli. Rather, more mundanely, an important foundation of rigorous empirical research of any kind is that the conclusions follow logically from the evidence. If such conclusions are kept modest and specific, then small and narrow datasets, appropriately analysed, can be a perfectly adequate form of empirical support. The problem comes about when the scope of the conclusions creep beyond what the collected data can support. This distinction is non-trivial in practice, despite sounding quite self-evident in theory.

What this means is that there is no general or simple answer to the question of “how big does my dataset need to be?”. The answer always depends on first answering “for what?”, and on answering several other questions, such as how the data are to be analysed, and how the end results of such analyses are to be linked to the theoretical claims of interest. Bigger is also not always better if size comes at a cost to data

quality or how meaningfully the data actually inform the research question being investigated. These facts more generally underscore how difficult it is to learn anything empirically, and resonate with perspectives that emphasize a balanced engagement with both quantitative and qualitative approaches to making sense of data (Meng, 2021; Tanweer et al., 2021; McElreath, 2020; Lundberg et al., 2021; Yarkoni, 2022; Pearl & Mackenzie, 2018).

While the above complexities preclude making prescriptions about how specific studies should be run, in general, collecting larger and more diverse datasets will tend to be beneficial in at least two main ways. First, larger datasets can help to make findings more robust and replicable, supporting cumulative progress and trustworthiness of research (although, critically, many factors underlie replicable research; more data alone is no guarantee of replicable results c.f., Smith & Little, 2018). Second, more diverse datasets can empirically support more diverse and expansive conclusions. This latter point is important because we should care about more than just the rigour and replicability of our results. It is easier to maintain existing data collection practices and just be more careful in making narrower and more specific interpretations. Doing this alone may increase rigour, and that would be a good thing, but if we fail to also study more diverse forms of music and more diverse people, this may leave music psychology producing knowledge that is mostly inert in the real-world, applying only in narrowly defined contexts, and further entrenching existing structural biases about who and what gets studied and therefore understood. Larger and more diverse datasets have the potential to both increase the rigour and breadth of our understanding in music psychology. To explain why, I now turn to the notion of sampling and its role in empirical research.

Sampling to generalize

Sampling is a foundational idea in empirical research. It conceptualises the act of creating a dataset as drawing a subset of observations from a population by some process. This subset, or sample, is then intended to represent the broader population to some degree and thus to provide a tractable way to learn something about phenomena that could never be exhaustively studied in their entirety. Sampling theory is then a collection of ideas and procedures for assessing whether a given sampling process is likely to produce a sample that is sufficiently representative in this way to allow conclusions about the population to be drawn.

Sampling theory tends to be more associated with quantitative research that use statistical techniques to quantify bias and to assess the precision of population-level estimates (Cochran, 1977; Thompson, 2012), and that use approaches like probability sampling and stratified sampling to produce reliable samples. But sampling theory is also an important part of qualitative research, where non-statistical subjective criteria are used to assess the depth, richness, and saturation of a given sample (Patton, 2015). For example, a researcher studying practise habits of professional musicians qualitatively via semi-structured interviews needs to make the sampling decision of how many musicians to interview. One approach would be to use ‘saturation sampling’: to keep interviewing new musicians from their chosen sample until nothing new from the interviews is being learned. Quantitative and qualitative approaches to sampling can also be combined (Shaffer, 2017).

Regardless of approach, sampling theory provides a powerful framework to move from finite observations (your data) to generalizations about a potentially infinite number of observations (your theory). But this power to derive generalizations from data is only licenced when certain criteria are met. First, a target population(s) that the generalizations pertain to must be defined. Second, observations must be sampled from this population in an unbiased way. And third, there must be sufficiently many samples from the target population.

Setting a target population Defining a target population is crucial as it sets the parameters of the study and the potential scope of the findings. The broader the claim, the wider the implied target population, increasing the empirical challenge of sampling. For instance, “these results imply that 95% of people enjoy listening to music” is a very general claim that implies a broad open-ended target population—all humans and all music. In contrast, “these results imply that 61% of Australian university students enjoy listening to electronic dance music” makes a more specific claim with a narrower and more targeted population—Australian university students and electronic dance music. This could be even further broken down: students

from a particular Australian university or all? Undergraduate or postgraduate students? What particular kind of electronic dance music? Enjoying this music in what context? How is ‘enjoyment’ to be measured? Different answers to each of such questions implies subtly different target populations and correspondingly subtly different claims that might be made from their study.

Whether to make broader or narrower claims is a matter of researcher goals and is shaped by disciplinary norms. Some research is more descriptive in nature, where the claims are more tightly tethered to the specific contexts of the collected data. For example, studies in ethnomusicology tend to be more descriptive in this way, where data are intended in the first instance to accurately describe particular contexts from particular perspectives. In contrast, studies in music psychology tend to be more inferential in running experiments to test general hypotheses which through being supported by data seek to explain phenomena far outside of the laboratory. The difference is not that descriptive research is in any way atheoretical, or uninterested in explaining broader phenomena in the world outside of their observations, but rather more that in descriptive work, the data are not serving *directly* to support a particular theory or conclusion, but instead informing theorising more indirectly in synthesis with other forms of evidence and reasoning. In inferential research this link between data and theory is direct, at least in its intention. It is important to emphasise here that neither descriptive nor inference research is superior to the other, nor is inferential research inherently more scientific. To the contrary, some have even suggested that psychology has verged at times on being unscientific in its over-reliance on inferential research at the expense of having a solid descriptive foundation on which to build the theories that it then tests (Rozin, 2001).

Another important distinction here is that the concept of a ‘target population’ is not confined to human populations but is an abstract idea encompassing the universe of potential observations pertinent to intended claims. For example, in music psychology research, the target population includes consideration of which kinds of musical stimuli participants might listen to, and what kinds of contexts this listening takes place in. In other words, the target population in such a case is not a population that exists in the real world, but a hypothetical population of observations one could make about particular people, particular musical stimuli, and particular listening contexts. This is important because musical experiences are not uniform, even within a single person. How somebody might experience an opera performance in a theater will be very different from how they experience a heavy metal concert from within a mosh-pit, and experiencing an opera from within a mosh-pit or a heavy metal performance in a theater will also likely entail very different kinds of experiences. These kinds of differences and subtleties motivate being precise about what target population one intends to sample from and generalize to.

Representative sampling To learn something about a target population empirically one must sample observations from it. However, not all samples are adequate as the conclusions drawn depend on the extent to which the samples at least approximately resemble the characteristics of the target population, albeit on a smaller scale¹. Or at least it must be possible to adjust for or account for any known biases. For example, if one wanted to understand the distribution of musical ability in a given population, it would be a bad idea, and an example of sampling bias, to only study the subset of people in that population who have had formal musical training. Doing so, without correcting for this bias, would likely lead to skewed conclusions that over-estimate the musical abilities of non-musically trained individuals.

While this may seem obvious, it is relatively easy for this sort of bias to sneak into research studies. Music psychologists often work at or near institutions that formally train musicians, and as such, the kinds of people who happen to be nearby, have some free time, and who might be more motivated to learn about music through participating in such a study, will tend to be more musically trained than the general population. Just by recruiting the people most convenient to study (this is known as ‘convenience sampling’), one can easily end up with conclusions that do not generalize to the broader population and actually be more of a reflection of how musically trained individuals perceive and experience music.

Exposure to musical training is not the only source of bias music psychologists ought to be weary of when using convenience sampling as a recruitment method. Sampling bias is a widely acknowledged problem in

¹Note: It is rarely possible to know for sure whether a given sample is truly ‘representative’ of its broader population. Instead, the focus is usually on assessing the merits of the sampling process and making a judgment about the likelihood that this process would produce a representative sample.

the behavioral and social sciences, where there is a systematic over-reliance on samples consisting of native English speakers (Blasi et al., 2022; Kidd & Garcia, 2022) from Western, Educated, Industrialized, Rich, and Democratic (WEIRD) populations (Henrich et al., 2010, 2022). These biased samples of people often then participate in artificial tasks with unnaturalistic stimuli (Box-Steffensmeier et al., 2022; Westfall et al., 2015), and the resulting data tends to get analysed in overly simplistic ways that mask these sources of heterogeneity (Yarkoni, 2022; Gelman, 2015). All of this has been argued to constitute a ‘generalisability crisis’ (Yarkoni, 2022).

In music, while there is a longer tradition of cross-cultural research in empirical fields like ethnomusicology (Nettl, 2005), subfields like music psychology continue to rely primarily on biased convenience samples from WEIRD populations to make general claims about musical minds (Jacoby, Margulis, et al., 2020; Stevens, 2012; Hilton & Mehr, 2022; Savage et al., 2021). The possibility that this is may be a problem is underscored by those exceptions finding notable cross-cultural differences (Balkwill & Thompson, 1999; Drake & El Heni, 2003; Krumhansl, 2000; Castellano et al., 1984), and this has motivated an growing body of research in recent years making comparisons across different culturally defined populations (Polak et al., 2018; London et al., 2017; McPherson et al., 2020; Jacoby & McDermott, 2017; Smit et al., 2022; Athanasopoulos et al., 2021; Anglada-Tort et al., 2023), as well as larger scale comparisons assessing broader patterns of cultural variation globally (Mehr et al., 2019; Hilton, Crowley-de Thierry, et al., 2022; Hilton, Moser, et al., 2022; Savage et al., 2015; Ozaki et al., 2023; Jacoby et al., 2024). This newer body of cross-cultural research has documented surprising results that contradict common assumptions, such as the lack of octave equivalence in pitch perception among the remote Tsimané people of the Bolivian Amazon (McDermott et al., 2016), lack of stereotypical emotional associations for harmonies (Smit et al., 2022; Athanasopoulos et al., 2021), or even the fact that a person’s native language can affect their abilities to perceive melodies and rhythms (Liu, Hilton, et al., 2023). Results like these emphasise the importance of taking a critical stance toward common assumptions, and actually empirically testing them.

Sample size With a target population identified, and a way to obtain a relatively unbiased sample from this population, sufficiently many samples from the population must be made in order for support robust conclusions. There are several interrelated reasons why sample size is important. One reason relates to the fact that populations of interest are often complex and heterogeneous in nature. If they weren’t, if each individual observation in a population were identical, and you knew this in advance, then you would only need a single observation to summarise the population. But this is never the case for real research questions, where the implied populations of interest almost invariably comprises of a multiplicity of interacting cause-effect relationships, where only some of these relationships are the focus of inquiry. Sorting through such real-world complexity needs a critical mass of observations to be able to distinguish signal from noise, regardless of whether an analysis is quantitative or qualitative.

Several other factors necessitate sufficiently large sample sizes, but are more specific to particular quantitative or qualitative analysis approaches. For instance, under classical frequentist statistics, a minimum sample size is required to ensure the validity of statistical inferences, as per the Central Limit Theorem, which states that sample means will approximate a normal distribution as sample size increases. The approximate normality of the sampling distribution, often assured by a sufficiently large sample size, is a critical prerequisite for conducting various frequentist statistical tests such as t-tests, ANOVAs, and other inferential procedures. In contrast, under Bayesian statistics, there’s no minimum sample size for a statistical inference to be technically valid; however, in practice, small sample sizes yield unreliable predictions and pose challenges when fitting complex models to the data (McElreath, 2020)². Similarly, qualitative methods do not require a minimum viable sample size to be technically valid, but it is crucial to ensure a sample size that adequately represents the diversity and depth of experiences, perceptions, or phenomena under investigation, since this enables a

²For those unfamiliar with the difference between frequentist and Bayesian statistics, here is a short (oversimplifying) explanation. Frequentist statistics is the most common statistical approach and it is a formal method for deciding whether statistical results (e.g., the average of a series of scores) are likely to have arisen by chance and thus whether they should be interpreted. This is typically done by checking whether the ‘p-value’ of a given statistical test is above or below some threshold (by convention, typically 0.05). By contrast, Bayesian statistics tends to be less dichotomous, usually focusing on quantifying the uncertainty of statistical results, and also provides a way to combine empirical data with subjective knowledge or expertise. For more detail with examples from music research, see Williamon et al. (2021)

rich, nuanced understanding of the research context and strengthens the credibility and transferability of findings (Patton, 2015).

Sample size here does not always refer to the number of people being studied, but rather is the number of observations from the relevant target population that one wishes to generalise over. For example, a case-study may aim to study a single person but collect data on them extensively over a long period of time. Studying a single person here is not a problem, provided that the results are not intended to directly apply to other people without first accounting for relevant differences. By contrast, many studies in music psychology do more directly intend to provide insight that generalises across different people and different musical stimuli. So correspondingly, it is important to sample sufficiently many people and stimuli from those populations of people and music the results are intended to generalise over (Hilton & Mehr, 2022).

Empirical musicology’s historical tendency to draw conclusions from small datasets is therefore a problem. But it is a problem shared with other empirical sciences, underscored by the “reproducibility crisis,” which highlights the inability to reproduce many prior empirical results in subsequent studies (Open Science Collaboration, 2015; Ioannidis, 2005). Many factors contribute to irreproducibility, but insufficiently large datasets is one important contributor (Smaldino & McElreath, 2016). The issue worsens when studying complex phenomena. The statistician and political scientist Andrew Gelman once suggested a rule of thumb that estimating an interaction effect (within a quantitative framework using linear regression) requires roughly 16 times the sample size needed for a simple main effect (Gelman, 2018). Yet, such realities are routinely ignored in practice. Thus, not only is it hard to change practices around sample sizes when studying simple phenomena (Smaldino & McElreath, 2016), but this gets even harder when studying real-world complexity and nuance (Gelman, 2015).

Interim summary Putting all of this together, many research questions in music psychology imply a much broader target population than what is actually sampled in practice. By failing to achieve sufficiently large and representative samples from their implied target populations, the logical bridge between data and conclusions crumbles away, compromising the reproducibility and generalisability of the conclusions drawn. The difficulty in remedying this situation is that scaling up sampling to obtain larger and more diverse datasets tends to be costly and hard to achieve in practice using traditional approaches. With this as a motivation, I now turn to web-based citizen science as one approach that is suited to the acquisition of larger and more diverse datasets compared to many other methods, and it does so in a relatively cost-effective manner (Hilton & Mehr, 2022).

How web-based citizen science can help

What is web-based citizen science? Web-based citizen science is an approach to data collection that relies on intrinsically motivated volunteers to contribute data via the internet to research projects. These two features of intrinsic motivation and internet-mediated participation are key to making this approach scalable for collecting large and diverse datasets.

Intrinsic motivation refers to the drive to engage in activities for their own sake, out of personal interest, enjoyment, or the inherent satisfaction derived from the activity itself. Citizen science studies are designed to align with and engage such intrinsic motivations as the primary means of recruiting people to participate in a study, negating the need for costly extrinsic incentives such as monetary compensation. This requires a more human-centered design approach that engages seriously with people’s motivations and that pays attention to people’s experiences while participating in a study. This contrasts with more traditional approaches to study design, which are under no obligation to be, and usually are not, either particularly interesting nor engaging for the participant.

Actually engaging and maintaining intrinsic motivation is important for citizen science studies because if a participant stops being motivated they simply stop participating. While participants in regular lab-based studies can also leave at any time, in doing so, they usually have to forfeit some amount of their extrinsic incentive, negotiate the social awkwardness of telling this to the experimenter, and of course, will have paid an opportunity cost of taking time out of their day to the commute into a lab for the experiment in the first

place. For a participant in a web-based citizen science study, they just close the app or browser tab on their device and think of it no more. As such, citizen science studies not only need to pitch an interesting topic to get participants through the front door—the initial ‘hook’ of a study—but must maintain engagement throughout. Design principles and ideas from gamification can be useful for maximizing engagement and enjoyment, to this aim (Long et al., 2023).

The benefits of intrinsic motivation is complemented by internet-mediated participation, widening the pool of potential participants from local to global by connecting with billions of people with internet-connected devices, at least, in principle. This can be achieved via purpose-built apps, or more simply by an interactive webpage. Participating via the internet not only widens the net of who can participate, but also makes participation much more convenient, since it can take place whenever and wherever it suits the participant. This contrasts sharply with traditional in-person approaches to data collection that require would-be participants to attend a particular location (e.g., a university lab) at a particular pre-arranged time (e.g., at 9am on a Tuesday).

Citizen science for scale The sheer magnitude of data obtained through citizen science can be astonishing, with some projects amassing data from millions of individuals worldwide, as opposed to merely tens of people from a local campus. For example, the Moral Machine Experiment engaged over 2 million demographically diverse individuals from 233 countries and territories to make judgments about ethical dilemmas involving autonomous vehicles (Awad et al., 2018). This scale allowed the authors to investigate how moral intuitions varied subtly across different cultural and demographic backgrounds globally, unlocking entirely new research questions that were hardly conceivable within the scope of traditional data collection methodologies. Similar approaches have also been applied to language acquisition (Hartshorne et al., 2018; Chen & Hartshorne, 2021), spatial navigation (Coutrot et al., 2022), personality (Bleidorn et al., 2016), and cognitive changes over the lifespan (Halberda et al., 2012; Hartshorne & Germine, 2015; Germine et al., 2012), among others.

Citizen science has a long history of enabling researchers to scale up data collection to obtain otherwise infeasible insights. This even predates the internet. Some of the first citizen science projects were in domains like conservation and ecology, where amateur naturalists would contribute observations of their local environments, allowing more comprehensive surveys of bird or plant species than professional scientists could achieve through their own fieldwork alone (Miller-Rushing et al., 2012). This tradition persists today with web-based citizen science projects like ‘eBird’, a popular app for logging bird sightings that boasts approximately half a million active users worldwide, records millions of bird observations, contributes to hundreds of scientific papers, and is regularly used in real-world conservation applications (Sullivan et al., 2014). Like with the Moral Machine Experiment, eBird allows researchers to harness millions of observations from around the world, making previously unattainable empirical questions requiring large and diverse datasets a practical reality (e.g., Freeman et al., 2022).

Web-based citizen science for music research Web-based citizen science has also been used to study music. One of the oldest web-based citizen science studies in music research is the game “Hooked On Music”, which was designed to test what makes music memorable and involved recruiting, to date, nearly 200,000 participants to listen to short excerpts of pop songs (Burgoyne et al., 2013; Honing, 2021). The game’s success in attracting many participants is in part because the experiment’s design mirrored a popular game that many people from western cultures already had familiarity with from TV quiz shows and school-yard games: trying to recognise a popular song from just a short excerpt.

More recently, a group of researchers (including the present author) at The Music Lab have released and published several research papers using citizen science as a data collection methodology. The first published example is the “World Music Quiz” which engaged 29,357 participants to listen to song recordings from diverse cultures around the world, and then guess whether each was, in its original context, a dance song, love song, healing song, or a lullaby song (Mehr et al., 2019)³. This game was designed to be appealing to participants by tapping into their curiosity about music from other cultures. This motivation was enhanced

³You can play the original game on this link - <https://www.themusiclab.org/quizzes/fc>

by deliberate design elements, such as, following each trial with a brief explanation akin to ‘liner notes’ that encapsulated information about the culture and context from which each song originated, and included a photo, usually of the performers of the music.

In a follow-up study, the “Kids’ Music Quiz”, this design was adapted to be appropriate and engaging for young children (with the help of their parents), surveying the same intuitions about the same recordings in 5,033 children between 3 and 16 years of age, and comparing their intuitions to those of 98,150 adults from the “World Music Quiz” (Hilton, Crowley-de Thierry, et al., 2022)⁴. An animated character ‘Susie the Star’ guided the child and their parent through the study and provided encouragement, and clear design elements communicated to the parent when they should be helping (e.g., filling out demographic information about the child’s age) and when they should not intervene (e.g., when the child is playing the game). The game was then widely promoted to families with young children on the website <https://childrenhelpingscience.com/>⁵ (c.f., Sheskin et al., 2020) and via social media. More generally, however, recruitment in such citizen-science studies can come in bursts, often related to specific events. Figure 1 shows recruitment of participants over time for both the “World Music Quiz” and the “Kids’ Music Quiz” over a span of roughly three years, with annotations highlighting when ‘bursts’ of recruitment coincided with specific events such as news coverage or viral social media engagement.

One of the unique design options unlocked by the larger sample sizes of web-based citizen science studies is exploring larger sets of stimuli than what is typical in most music psychology experiments. The “Who’s Listening?” game set out to take advantage of this, using a corpus of 1,615 vocal recordings from 21 diverse societies as stimuli. The game consists of a series of trials in which participants listen to these recordings and guess whether the person in the recording was vocalising to an adult or to an infant⁶. This game attracted 51,065 participants from 187 countries, natively speaking 199 languages, resulting in hundreds of ratings for each of the 1,615 recordings (Hilton, Moser, et al., 2022). This combination of many people and many stimuli is uniquely powerful for addressing questions about generalisability because it allows robustly quantifying variation across people, across stimuli, and addressing questions of how the two interact (e.g., what is the effect of a recording being from a similar culture to that of the listener).

One of the most popular games on The Music Lab website is called “Test Your Musical IQ”⁷. It has attracted nearly a million participants worldwide to complete in a 20-minute-long test of musical perception abilities, and is comprised of subtests for melodic perception (Harrison et al., 2017), mistuning perception (Larrouy-Maestri et al., 2019), and beat perception (Harrison & Müllensiefen, 2018). Several factors likely combined to make this game so popular. First, the framing of the game, encapsulated by its title, presents a simple proposition that leverages the salience of notions of ‘IQ testing’ in popular culture—inviting the participant to complete a test, and then receive a score that allows them to compare their abilities to others. Inadvertently, the controversial nature of IQ testing, and its various misuses, may have also driven even more attention to the game, with popular music YouTubers such as Adam Neely posting video critiques. Secondly, the design of the game encouraged such viral social media engagement by providing participants with an attractive data visualisation of their results on the test, along with easy ways to share it on social media platforms. This helps to facilitate ‘snowballing’ of participant recruitment, such that each participant in turn helps to recruit their friends to be participants too.

What can a dataset of this massive scale enable? Liu, Hilton, and colleagues (Liu, Hilton, et al., 2023) sought to test the hypothesis that speakers of tonal languages (e.g., Mandarin or Yoruba) are better at perceiving musical melodies compared to speakers of non-tonal languages (e.g., English or Bengali). Previous studies that had tested this hypothesis had largely compared native English speakers from the United States to native Mandarin speakers from China. Using a subset of the ‘Test Your Music IQ’ data, involving 459,066 participants from 203 countries and territories, they were able to test this theory by comparing between native speakers of 19 tonal languages to 35 non-tonal languages, with each language being represented by people from many countries. The result not only confirmed the robust generality of the boost to melody

⁴If you have children, you can play this game with them here - <https://www.themusiclab.org/quizzes/kfc>

⁵Children Helping Science is a collaborative organisation specifically aiming to support the recruitment of children and families in online developmental psychology studies

⁶You can play the original game here - <https://www.themusiclab.org/quizzes/ids>

⁷Play the game here - <https://www.themusiclab.org/quizzes/miq>

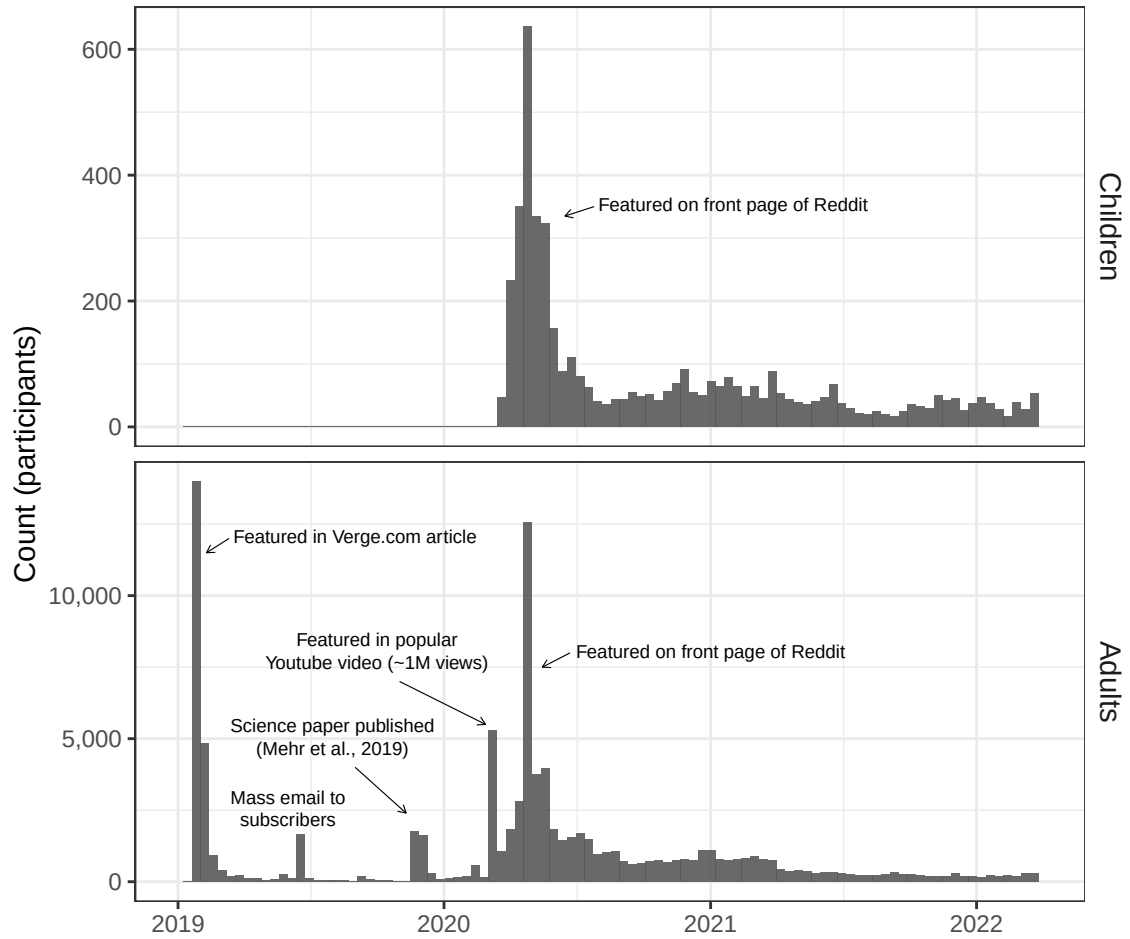


Figure 1. | The time-series of participant recruitment demonstrating spikes corresponding to specific events The histograms show the number of participants (after exclusion) that completed the the ‘World Music Quiz’ (bottom) and the ‘Kids Music Quiz’ (top) between 2019 and 2022. The date of participation is indicated on the x -axis. The annotations on each plot indicate specific events that coincided with spikes in participation, such as viral posts on social media, news coverage, etc.

perception among tonal language speakers, but surprisingly, found that this reliably came with a tradeoff, where tonal language speakers were on average worse at beat perception compared to speakers of non-tonal languages, a finding that proved consistent across global variation in age, linguistic background, cultural upbringing, educational level, socioeconomic status, and musical expertise.

Limitations and challenges Web-based citizen science can be an effective way to scale up data collection to achieve larger and more diverse samples of participants in music psychology experiments. In many cases, it is able to achieve this to such an extent that it would be unthinkable to match this scale via more traditional in-person laboratory methods. But there are also several limitations of this approach that should be considered.

First, while web-based citizen science will tend to recruit more diverse participants than typical lab-based studies, this does not mean that citizen science samples are without bias, or that citizen science is an effective way to representatively sample a specific target population. Most obviously, only people with internet connected devices have even the possibility of participating. While an increasingly large portion of the global population is connected to the internet, there are still important populations that have limited or no internet access. And indeed, it is often precisely these populations that will be less influenced by globalising cultural influences, and thus their data, inaccessible to web-based citizen science methods, may be particularly theoretically informative for hypotheses about human psychological variation generally (Majid, 2023). For these reasons, other recruitment methodologies, such as field-work, and other forms of more targeted sampling, will remain important as ever.

Secondly, even restricted to those with internet access, citizen science is itself a form of convenience sampling and does not guarantee unbiased samples of participants. In particular, citizen science relies on intrinsic motivation to drive recruitment, and people have different interests. Those whose interests align with a given study will be more likely to participate than those with other interests. This can lead to sampling bias. For example, citizen science studies about music will likely be particularly appealing to those with stronger interests or hobbies involving music. Depending on a project’s research question, this is not necessarily a problem. But for example it would pose serious issues if one aimed to estimate the prevalence of passionate music fans in the general population—this would be an example of what is sometimes called ‘self-selection bias’.

One of the more effective ways to mitigate sampling bias that stems from differences in intrinsic motivations is to ensure that the study is fun and accessible in as general a way as possible. That is, if participation involves hours of tedious activities, then only the most passionate music fans with interest in science and free time to participate will be likely to participate to completion—exacerbating biases. By contrast, short and enjoyable activities effectively lowers the bar for how much intrinsic motivation a person needs to participate, making it more likely that a greater number of more diverse people will participate. This is an important design consideration that needs to be considered along with the goals of a study, but it is also an important limitation, in that some research questions may simply require more laborious and longwinded tasks, and therefore be less suited to citizen science style recruitment.

Other biases may also apply. For example, younger people tend to be more active on the internet and therefore more likely to participate in studies on the internet. Biases like these, however, are relatively easy to statistically correct. This is because it is easy to measure the age of a participant through a self-report question. So if one is concerned that an unrepresentative portion of young people might skew an analysis, then stratifying by that person’s self-report age using linear regression can mitigate this concern (i.e., including age as a predictor in a model). A similar approach may also be taken to account for imbalances in the number of participants from different countries, or speakers of different languages, although in these cases, the use of multi-level models can be a more elegant solution (Hilton & Mehr, 2022; Liu, Hilton, et al., 2023; Hilton, Moser, et al., 2022)⁸. Demographic variables in a sample can even be statistically adjusted in more targeted ways to allow more direct tests of generalisability to different populations of interest (Deffner

⁸Multi-level, or ‘mixed-effects’, regression models provide a way to estimate overall statistical effects that are not biased by imbalances in sample size across different subgroups in the data, and also tend to provide better estimates of subgroups with smaller sample sizes by leveraging information in the other groups. For a recent review of this approach see Hoffman & Walters (2022).

et al., 2022). Sophisticated analysis approaches like these become more feasible and more effective at the scale of data made possible by citizen science samples.

A notable difficulty of web-based citizen science studies is that to implement them, one generally requires some degree of web development and computer programming skills, and these skills are rarely part of standard researcher training. Mitigating these difficulties, however, a growing number of tools make implementing web-based studies easier, such as jsPsych (de Leeuw, 2023), Data Pipe (De Leeuw et al., 2023), Pushkin (Hartshorne et al., 2019), PsychoPy (Peirce et al., 2019), among other currently in-progress efforts. Basic skills with programming languages like javascript, HTML, and CSS are also increasingly accessible with abundant free or cheap resources. So while these implementation challenges are real, they are not insurmountable, especially to a generation of early-career researchers with time to learn new and valuable skills.

In addition to these technical challenges, citizen science studies also require a fundamentally different design approach to that typically taken in lab-based psychology studies. Just transplanting a typical experiment from the lab to the internet will rarely be effective since typical experiments are simply not designed to be interesting or engaging. More often they are excessively repetitive, aesthetically uninspired, and just plain boring. Instead, effective citizen science studies must balance scientific criteria against human motivational factors, and it is far from trivial to get this balance right in practice: just because a design is engaging does not mean it will collect meaningful data, and vice versa. An important part of this, and a key difference to the way most psychology experiments get designed, is not just thinking through the logic of various design decision but rather engaging in loops of ‘iterative design’ where prototypes are extensively ‘play tested’ with real participants as a means of assessing what elements of a given design are clear and engaging and which are not. These efforts do not need to reinvent the wheel, since borrowing ideas from ‘gamification’ and game-design more broadly offer a lot of largely-untapped wisdom and points of inspiration (Long et al., 2023).

Conclusion

In this chapter, I have argued that researchers in music psychology should collect larger and more diverse datasets to empirically support more robust and generalisable findings. This recommendation resonates with the notion of a rigorous treatment of data-rich research questions that was at the heart of Clarke and Cook’s original conception of empirical musicology (Clarke & Cook, 2004). More specifically to music psychology, this recommendation comes out of a growing discussion around the need to study musical minds more diversely and the various challenges inherent to expanding our empirical scope in this way (Jacoby, Margulis, et al., 2020). One of these challenges is the logistical challenge of how to feasibly collect larger and more diverse datasets and this is what I have focused on in this chapter, first, by outlining the conceptual foundation of why we need larger and more diverse datasets, and second, by suggesting web-based citizen science as an approach uniquely suited to economically scaling up data collection.

This web-based citizen science approach has proven effective in several recent studies but is not free from limitations and challenges and is not the only methodology that can help with scaling up the size and diversity of datasets. Another complementary approach is to scale up by involving more collaborators: the so-called ‘many labs’ approach (Wang, 2021; Van Essen et al., 2012; Klein & et al., 2014; Klein et al., 2018; Consortium, 2020). By pooling resources from multiple researchers for a single project, this approach can improve the ability to collect large samples, and has been effective for some empirical projects studying music (Hilton, Moser, et al., 2022; Yurdum et al., 2023; Ozaki et al., 2023). Although the diversity of samples obtained by such an approach is limited by the diversity of researchers in the team (Schimmelpfennig et al., 2023), so it is also advisable to think about diversity and inclusion in terms of broader questions of who participates in the research (Savage et al., 2023). Indeed, increasing the diversity of who participates in research is also important because biases in data are just as important as biases in the theories we approach such data with (Ewell, 2020).

In conclusion, to produce more nuanced, generalisable, and cross-culturally representative understandings of musical minds, music psychology needs to collect larger and more diverse datasets which can uniquely be facilitated by a web-based citizen science approach.

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